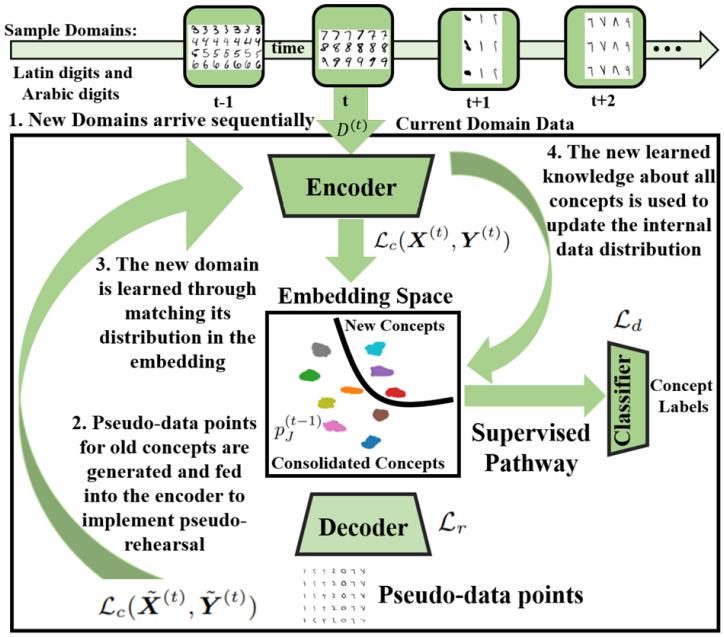
COMSPCI 713 Part 2 – Mock questions and answers

Question 1 [4 marks]

The paper *"Cognitively Inspired Learning of Incremental Drifting Concepts"* by Rostami & Galstyan introduces a continual learning system designed to handle concept drift. The diagram below illustrates the core steps of this method:



Incremental Concept Learning Process

Using your knowledge of continual learning from the lecture, answer the following questions:

1) What specific challenge in continual learning is this approach designed to address? [1 mark]

This method addresses the challenge of concept drift in continual learning, where data distributions change over time. It specifically aims to prevent catastrophic forgetting—the tendency of neural networks to forget previously learned information when trained on new tasks—while incrementally learning new concepts from sequentially arriving domains.

(1 mark for identifying forgetting and or concept drift and explaining briefly what it is)

2) Step 2 in the diagram shows the generation of pseudo-data points. Which of the three major categories of continual learning strategies covered in class does this step belong to? Briefly describe the key idea behind this category. [2 marks]

Step 2 belongs to Replay-based methods. Replay-based approaches address forgetting by revisiting information from past tasks during training on new ones. This is done by storing or generating past examples (e.g. pseudo-data), which are replayed to the model to maintain performance on earlier tasks.

(1 mark for identifying replay-based methods, 1 mark for brief description)

3) The approach uses a Gaussian Mixture Model (GMM). What is the purpose of the GMM in this system? [1 mark]

The GMM models the distribution of previous concepts in the embedding space. It enables the generation of representative pseudo-data points for old tasks, supporting pseudo-rehearsal without storing real data and helping mitigate catastrophic forgetting.

(1 mark for identifying role of GMM and briefly explaining that it models distribution of concepts and is used to sample pseudo-data points)

Question 2 [3 marks]

Uniform Cost Search (UCS) and Breadth-First Search (BFS) are two commonly used tree search algorithms.

1) Describe how each algorithm explores a search tree. [1 mark]

Breadth-First Search (BFS): Explores the search tree level by level, expanding all nodes at a given depth before moving to the next level. (0.5 mark)

Uniform Cost Search (UCS): Expands the node with the lowest total path cost from the root, regardless of depth. (0.5 mark)

2) Suppose the edges in a search tree have different costs. For each algorithm, state whether it is optimal in this case and explain why. [2 marks]

BFS is not optimal when edge costs differ, because it prioritizes shallow nodes without considering cost. (1 mark)

UCS is optimal for varying edge costs (if all edge costs are positive), as it always expands the least-cost path first, ensuring the lowest-cost solution is found. (1 mark)

Question 3 [3 marks]

Monte Carlo Tree Search (MCTS) can be implemented using an Upper Confidence Tree (UCT), which relies on the Upper Confidence Bound (UCB1) formula shown below:

$$UCB1(n) = \frac{U(n)}{N(n)} + C \times \sqrt{\frac{\log N(\text{PARENT}(n))}{N(n)}}$$

Where:

- $U(n)$: total utility (reward) from simulations passing through node n
- $N(n)$: number of times node n has been visited

Briefly explain the role of each of the following components in the formula:

1. $\frac{U(n)}{N(n)}$
2. C
3. $\sqrt{\frac{log(N(PARENT(n))}{N(n)}}$

4. $\frac{U(n)}{N(n)}$

This is the exploitation term. It represents the average reward obtained from choosing node n, encouraging the selection of nodes with high observed utility (i.e., leading to high rewards). (1 mark)

5. C

This is a parameter (tunable) that balances exploration and exploitation. A higher value of C increases the preference for exploration. (1 mark)

6. $\sqrt{\frac{log(N(PARENT(n))}{N(n)}}$

This is the exploration term. It gives higher value to nodes that have been visited less often, encouraging exploration of less tried options. It grows with the parent's visits and shrinks as the node is explored more. (1 mark)

Question 4 [1 mark]

In a simplified Pac-Man environment, the agent must eat all dots in a maze as efficiently as possible, while avoiding moving ghosts. Pac-Man can move in four directions and sees nearby tiles. The goal is to clear the maze efficiently without getting caught by the ghosts.

Which option best defines the **State space**, **Action space**, **Policy**, and **Reward** of a Reinforcement Learning scenario? [1 mark for selecting the correct answer, no explanation requires]

A)

- **State space:** Pac-Man's position, nearby dot positions and ghost positions
- **Action space:** {up, down, left, right}
- **Policy:** Mapping of Pacman's position to action probabilities
- **Reward:** +10 for eating a dot, -100 if caught by a ghost, -1 per move

B)

- **State space:** Pac-Man's position, nearby dot positions and ghost positions
- **Action space:** {up, down, left, right}
- **Policy:** Mapping of the observed state to action probabilities
- **Reward:** +10 for eating a dot, -1 per move

C)

- **State space:** Pac-Man's position, nearby dot positions and ghost positions
- **Action space:** {up, down, left, right}

- **Policy:** Mapping of the observed state to action probabilities
- **Reward:** +10 for eating a dot, -100 if caught by a ghost, -1 per move

D)

- **State space:** Pac-Man's position only
- **Action space:** {up, down, left, right}
- **Policy:** Deterministic path from start to finish
- **Reward:** +1 per move, +100 when all dots eaten

Correct answer: C (use Ed Discussion to discuss why) (1 mark for selecting the correct answer, no explanation needed)

For this kind of question, it can be useful to process by elimination.

Question 5 [2 marks]

In the context of Graph Neural Networks (GNNs), briefly explain the concepts of permutation invariance and permutation equivariance.

In GNNs, permutation invariance means that the final graph-level output (e.g., for classification) does not depend on the order of the input nodes — the output remains the same regardless of how nodes are ordered.

Permutation equivariance means that if the nodes in the input graph are reordered, the node-level outputs are reordered in the same way. This is important for preserving the structure during message passing.

(1 mark if each explanation makes sense, and is associated to the correct term)

Question 6 [2 marks]

In the context of Self-Supervised Learning (SSL):

1. What is the difference between a pretext task and a downstream task? Provide a brief explanation. [1 mark]

The pre-text task is an auxiliary task designed to train the model to learn useful representations from unlabelled data (e.g., predicting missing parts of input). (0.5 mark)

The downstream task is the real-world task of interest (e.g., image classification), where these learned representations are applied. (0.5 mark)

2. When evaluating a self-supervised learning system, which task's performance is more important, and why? [1 mark]

You should care most about performance on the downstream task, because it reflects how well the learned representations generalise to practical applications — the true goal of SSL.

(0.5 mark for identifying downstream task classification is more important and 0.5 mark for explaining why)